BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 102101

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. If P(x) and Q(x) are two polynomials then P(x)/Q(x), Q(x) ≠ 0 is called _____.

(A) Proper fraction
(B) Irrational fraction
(C) Improper fraction
(D) Rational fraction

Q2. A chord 8√3 units long is 4 units from the centre of a circle. The length of the radius of the circle is _____.

(A) 14.4 units
(B) 8 units
(C) 8√2 units
(D) 2√8 units

Q3. The measure of an angle in a semi-circle is equal to _____.

(A) 45°
(B) 60°
(C) 90°
(D) 180°

Q4. AB and CD are two chords of the same circle and AB < CD, then

(A) AB is closer to zero
(B) AB must be parallel to CD
(C) Can't decide
(D) CD is closer to zero

Q5. Two chords of a circle which are equidistant from the centre are

(A) Congruent
(B) Parallel
(C) Not congruent
(D) Not parallel

Q6. The difference between upper limit of two consecutive classes in a frequency table is called _____.

(A) Class limit
(B) Class interval
(C) Class mark
(D) Range

Q7. 45° = _____ radians

(A) π/3
(B) π/2
(C) π/6
(D) π/4

Q8. $1+Cot^2\theta$ = _____.

(A) $Cosec^2\theta$
(B) $Cos^2\theta$
(C) $Sin^2\theta$
(D) $Sec^2\theta$

Q9. Identify the item that does not have the same ratio as the other three.

(A) 30/45
(B) 4 to 6
(C) 2:3
(D) 3 to 2

Q10. If $m \propto 1/n$ then.

(A) mn = k
(B) m/n = k
(C) n : km
(D) m = kn

Q11. What are the solutions of the equation $x^2+7x-18=0$.

(A) -2 or 9
(B) 2 or -9
(C) 2 or 9
(D) -2 or -9

Q12. 1/w = _____.

(A) w^3
(B) w
(C) w^2
(D) None of these

Q13. The sum of the roots of the equation $x^2-5kx+6k^2=0$ is _____.

(A) -5k
(B) $5k^2$
(C) -5/6k
(D) 5k

Q14. The equation $x^4-5x^2+4=0$ is called _____ equation.

(A) Quadratic
(B) Cubic
(C) Quartic
(D) None of these

Q15. $(3x-2)/(2x^2-x)$ = _____.

(A) 2/x + 1/(2x-1)
(B) 2/x - 1/(2x-1)
(C) 1/(2x-1) - 2/x
(D) None of these

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 36

QII. Attempt any NINE Parts out of the following. Each Part carries equal marks.

(i) For what value of k the roots of the equation $12x^2+kx+3=0$ are equal.

(ii) Evaluate: $(1+3w - w^2)(1+w- 2w^2)$.

(iii) Solve the quadratic equation $9x^2-6x + 4/9=0$ by completing the square.

(iv) There is a direct variation between x^2 and y when x=7, y=49, Find y when x=9.

(v) If α , β are the roots of $3x^2-2x+5=0$, find the equation whose roots are α/β , β/α .

(vi) If x = {1,2,3,4} and y = {5,6,7,8}, then write a one-one function from x to y.

(vii) From the following distribution Daily Wages (in Rs) : 112–116 117–121 122–126 127–131 132–136 No. of workers : 3 20 12 7 5 Calculate Harmonic Mean.

(viii) Resolve into Partial fraction: $(x+32)/((x+2)(x^2+5x+6))$.

(ix) Verify De Morgan's law for the following sets. U={1,2,3,...,10} , B={1,3,5,7,9} , A={2,4,6,8,10}

(x) In a fountain, the pressure "P" of water at any internal point varies directly as depth "d". From surface pressure is 51 N / cm^2 when depth is 3cm. Find pressure when depth is 7cm.

(xi) A tree is 50 m high. Find the angle of elevation of its top to a point, on the ground 100m away from the foot of the tree.

(xii) Prove that $Cos^4b - Sin^4b = 1-2 Sin^2b$.

SECTION "C"

Maximum Marks: 24

> Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII. If two chords of a circle are congruent then they will be equidistant from the centre.

QIV. If a line is drawn perpendicular to a radial segment of a circle as its outer end point, it is tangent to the circle at the point.

QV. The angle in a segment less than a semi-circle is greater than a right angle.

QVI. Construct a triangle ABC such that mAB=3" mAC=4" and m∠A =60°, Draw circumcircle to this triangle.